Advanced IP Analyser

Instruction Book

VERSION 2.0.0

Debian GNU/Linux 13 (Trixie)

Scanning - Packet Analysis - Network Watch - Passive Wi-Fi

Contents

Use this guide for installation, daily operation, packet analysis, monitoring, updates, and troubleshooting.

About this book

This book explains how to install, operate, update, troubleshoot, and safely administer Advanced IP Analyser. It covers the graphical application, CLI, packet display filters, Network Watch, and Passive Wi-Fi Watch.

Advanced IP Analyser is intended only for networks, devices, and radio traffic you own or are explicitly authorized to inspect. The software does not grant authorization and does not determine whether a particular capture is lawful.

1. What Advanced IP Analyser does

Advanced IP Analyser combines four workflows:

send Wake-on-LAN, and perform confirmed SSH power actions.

inspect metadata/bytes, and apply Wireshark-style display filters.

Wi-Fi monitor-mode traffic.

1. **Inventory:** find reachable IPv4/IPv6 devices and open TCP services.
2. **Administration:** save devices, exchange inventories, launch services,
3. **Packet analysis:** capture a bounded host/service scope, open packet files,
4. **Monitoring:** analyze traffic over time and passively observe compatible

It is a focused Debian administration utility, not a complete protocol-forensics suite. It does not decrypt TLS, reconstruct every application stream, exploit a service, disconnect Wi-Fi clients, inject frames, or recover passwords.

2. System requirements

Required

- Debian GNU/Linux 13 (Trixie), with a graphical desktop for the GUI.
- Python 3.11 or newer and Tk.
- `iproute2`, `iputils-ping`, `xdg-utils`, `pkexec`, and `python3-defusedxml`.
- Authorization for every network and device placed in scope.

Optional

- `iw` and a compatible wireless adapter for Passive Wi-Fi Watch.
- `openssh-client` for SSH and remote power.
- `freerdp3-x11` for RDP.
- `gvfs-backends` for SMB browsing.
- `iputils-tracepath` and `telnet` for those diagnostic launchers.
- Debian IEEE or Nmap vendor data for offline MAC manufacturer names.
- `notify-send` for Network Watch desktop alerts.

3. Install and remove

Install the release package

Download `advanced-ip-analyser_2.0.0_all.deb` from the project release page. In a terminal:

```
cd ~/Downloads
sudo apt install ./advanced-ip-analyser_2.0.0_all.deb
```

Use `apt install ./file.deb`, not `dpkg -i`, because APT resolves dependencies.

Launch from the desktop menu or run:

```
advanced-ip-analyser-gui
```

The CLI executable is `advanced-ip-analyser`.

Remove

```
sudo apt remove advanced-ip-analyser
```

APT removes application files but normally leaves per-user favorites, captures, rules, history, bookmarks, and reports. Review the file locations in chapter 16 before manually removing personal data.

4. Main window tour

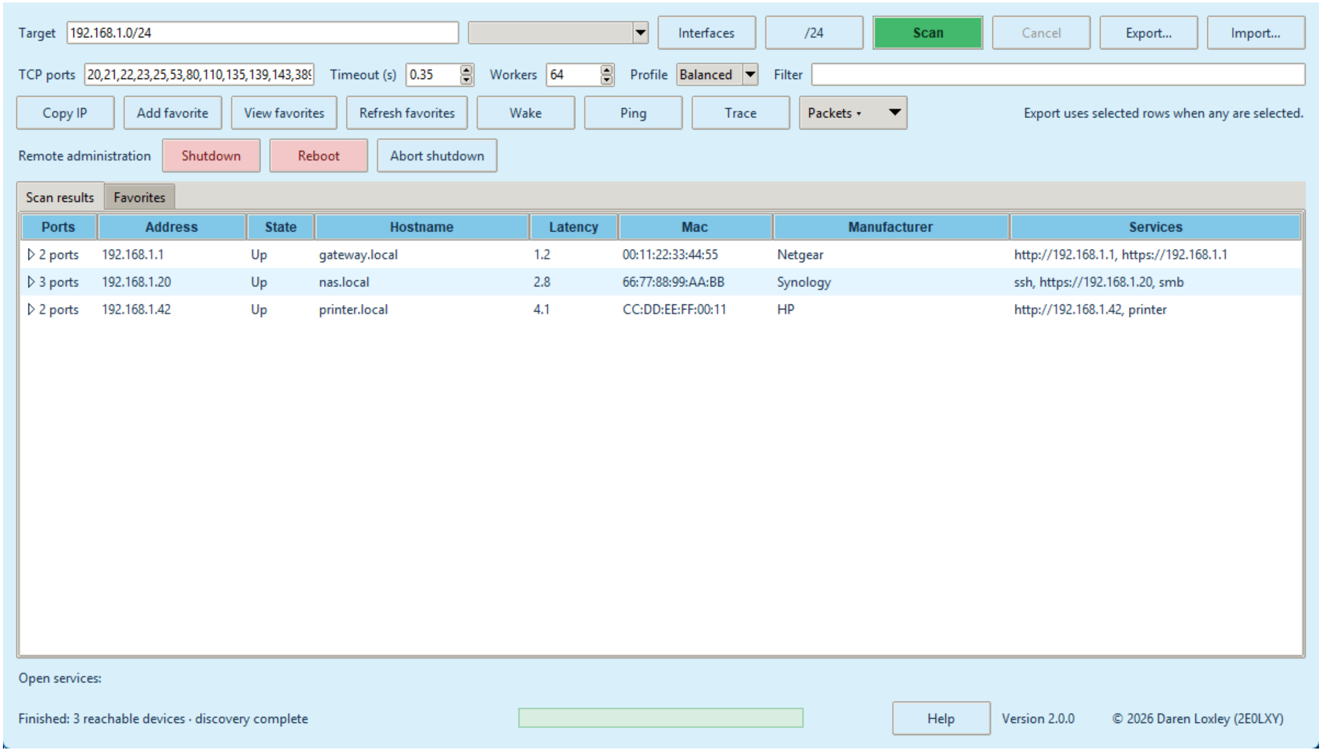

Figure: Scanner overview

The main window is arranged from setup to action:

- **Target:** an address, inclusive range, or CIDR.
- **Interfaces:** refresh active IPv4 interface/subnet presets.
- **/24:** use the current host's class-C-style subnet.
- **TCP ports:** comma-separated ports and ranges.
- **Timeout and Workers:** tune connection patience and concurrency.
- **Profile:** Fast, Balanced, or Accurate presets.
- **Filter:** live result-table text search.
- **Action row:** copy, favorites, Wake-on-LAN, Ping, Trace, and packet tools.
- **Remote administration:** Shutdown, Reboot, and Abort shutdown.
- **Scan results/Favorites:** inventory tables.

- **Footer:** status, centered green progress, Help, version, and Update.

5. Run a scan

Choose a target

Accepted examples:

```
192.168.1.20
192.168.1.10-192.168.1.50
192.168.1.0/24
2001:db8::10
2001:db8::/120
```

IPv4 CIDRs exclude network and broadcast addresses where applicable. Each scan is limited to 65,536 expanded addresses.

Choose ports and a profile

The default ports cover common web, file, mail, remote-access, database, printer, and management services. Enter values such as:

```
22,53,80,443,445,8000-8100
```

Right-click the port field for presets. An all-65,535-port scan is allowed only after confirmation. Use it on one or a few targets, not a whole subnet.

- **Fast:** short timeout and higher concurrency.
- **Balanced:** suitable for normal local networks.
- **Accurate:** longer timeout and lower concurrency for slow/filtering networks.

Start, follow, or cancel

Press **Scan** or `F5`. The first phase tests reachability and ports. Reachable devices appear as they are discovered. The second phase inspects safe server metadata for hosts with open ports. Its separate indicator prevents fingerprint work from delaying the initial inventory.

Press **Cancel** or `Escape` to retain partial results and stop pending work.

6. Understand and use results

The host row shows address, state, hostname, latency, MAC, manufacturer, and services. Click a heading to sort ascending; click again for descending.

Expand a host to see detected ports. Expand a port to see metadata such as:

- HTTP status, server/powered-by, content type, title, redirect, and realm;
- TLS protocol and cipher; or
- a bounded safe SSH, FTP, SMTP, POP3, or IMAP greeting.

Use the Filter field to search all visible host/service metadata. This is a table search; packet display filters are a separate language described in chapter 11.

Open a supported service by double-clicking, pressing Enter, using the context menu, or selecting a generated link. HTTP(S)/FTP use the desktop URL handler; SMB uses the file manager; SSH/Telnet open a terminal; RDP uses FreeRDP.

7. Favorites and inventory exchange

Select host rows and press **Add favorite**. Favorites are identified by MAC first, so a refreshed DHCP address does not lose the saved identity or note.

The Favorites tab supports:

- refresh/rescan;
- note editing;
- removal;
- export selected; and
- the same host actions available in scan results.

Export supports CSV, JSON, XML, and escaped HTML. When rows are selected, only those rows are exported; otherwise all visible rows are exported. CSV text that could become a spreadsheet formula is neutralized.

Import accepts only bounded Advanced IP Analyser JSON/XML. XML parsing blocks entities/DOCTYPE and uses a hardened parser. Imported observations merge into the visible inventory and favorites.

8. Wake, diagnostics, and remote services

Wake-on-LAN

Select devices with MAC addresses and press **Wake**. The app selects a matching active-interface broadcast where possible. Wake-on-LAN must also be enabled in the target firmware/OS and may not cross routers.

Ping and Tracepath

Select a host and use **Ping** or **Trace**. A Debian terminal opens with a fixed, bounded command. No shell command is built from network data.

Remote power

Shutdown and reboot use non-interactive OpenSSH and ask for confirmation. They schedule the remote action one minute ahead so **Abort shutdown** can cancel it. Configure SSH keys and narrowly scoped passwordless remote `sudo shutdown` permission. The application never requests, retains, or forwards passwords.

9. Capture selected traffic

Select one or more host rows and choose **Packets** → **Capture selected host/service**. Selecting one service row also restricts the capture to that port.

Choose a Linux interface (`any` is allowed) and 1-300 seconds. Live captures are limited to 100,000 packets. The GUI stays unprivileged; when raw sockets are denied, PolicyKit starts the installed root-owned helper. The helper validates interface, addresses, port, duration, packet limit, output ownership, file type, link count, and link type before writing.

The capture opens automatically in the packet viewer and can be saved elsewhere.

10. Open and inspect packet files

Use **Packets** → **Open capture file for selection**. If hosts or a service are selected, the reader restricts results to that scope. It accepts:

- classic little/big-endian PCAP with micro/nanosecond timestamps;
- PCAPNG Ethernet interfaces; and

- gzip-compressed PCAP/PCAPNG.

Expanded input is limited to 512 MiB, 100,000 packets, and 262,144 bytes per record. Unsupported or malformed structures are rejected.

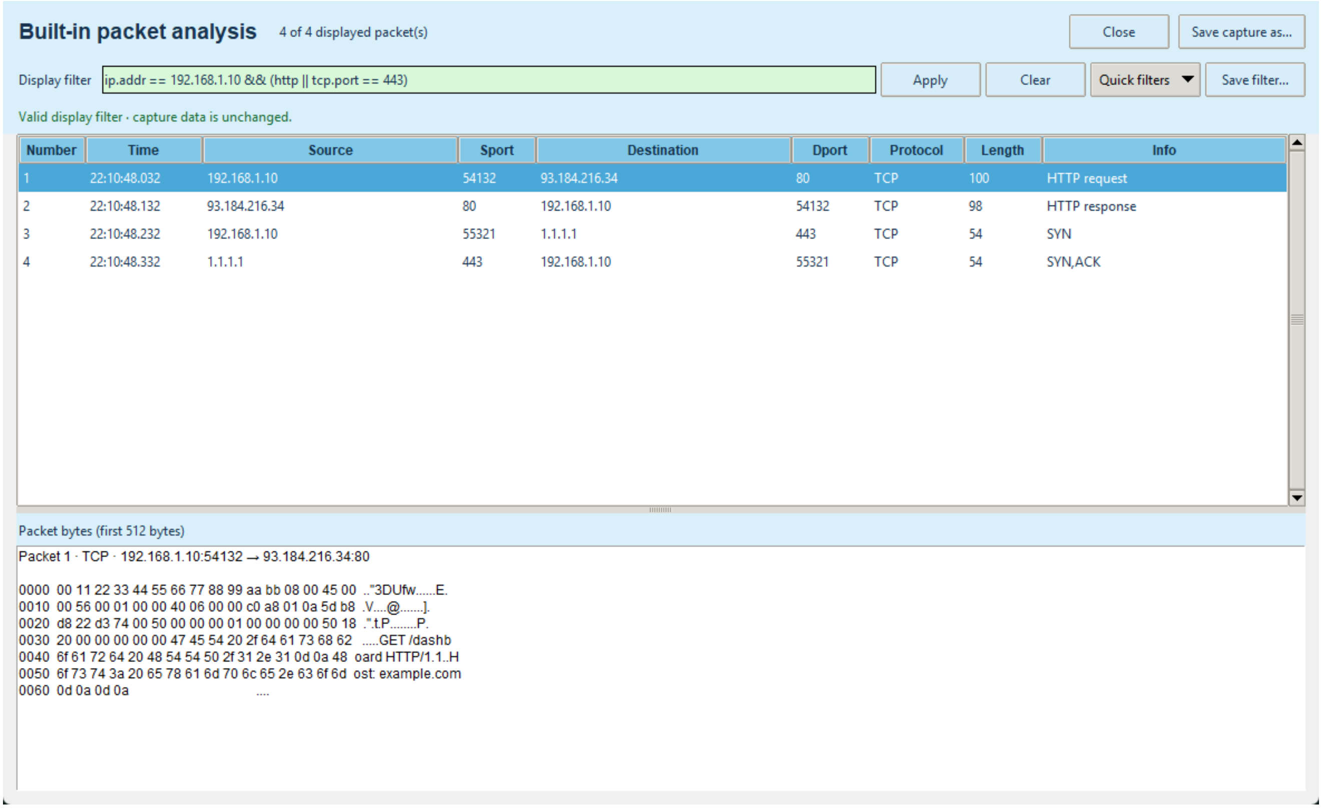

Figure: Packet display filters

Select a packet to inspect its summary and bounded hex/ASCII preview.

11. Display filters

Display filters change which existing packets appear. They never change what is captured. Press Enter or **Apply**. Green means valid; red provides an error.

Protocols and address fields

```
ip
ipv6
tcp
udp
dns
http
http.request
http.response
https
tls
icmp || icmpv6
arp
ip.addr == 192.168.1.10
ip.src == 192.168.1.10
ip.dst == 8.8.8.8
ip.addr == 192.168.1.0/24
```

Ports and TCP flags

```
tcp.port == 443
tcp.srcport == 443
tcp.dstport == 22
udp.port == 53
tcp.flags.syn == 1
tcp.flags.syn == 1 && tcp.flags.ack == 0
tcp.flags.fin == 1 || tcp.flags.rst == 1
tcp.len > 0
```

DNS, HTTP, and TLS

```
dns.flags.response == 0
dns.flags.response == 1
dns.qry.name contains "example.com"
http.request.method == "GET"
http.host contains "example.com"
http.request.uri contains "/login"
http.response.code == 404
tls.handshake.type == 1
ssl.handshake.type == 1
```

Size, text, and combinations

```
frame.len > 1000
protocol != "arp"
dns.qry.name matches "^api\\.example\\.com$"
ip.addr == 192.168.1.10 && (http || dns)
!(ip.addr == 192.168.1.0/24)
```

Comparison operators are `==`, `!=`, `>`, `<`, `>=`, `<=`, `contains`, and `matches`. Boolean operators are `&&`, `||`, `!`, and parentheses. Expressions, tokens, nesting, text values, and regex features are bounded. Regex lookaround, backreferences, groups, counted repetition, and heavy repetition are rejected.

The Quick filters menu provides 20 presets. Use **Save filter** to persist a validated named expression.

12. Network Watch

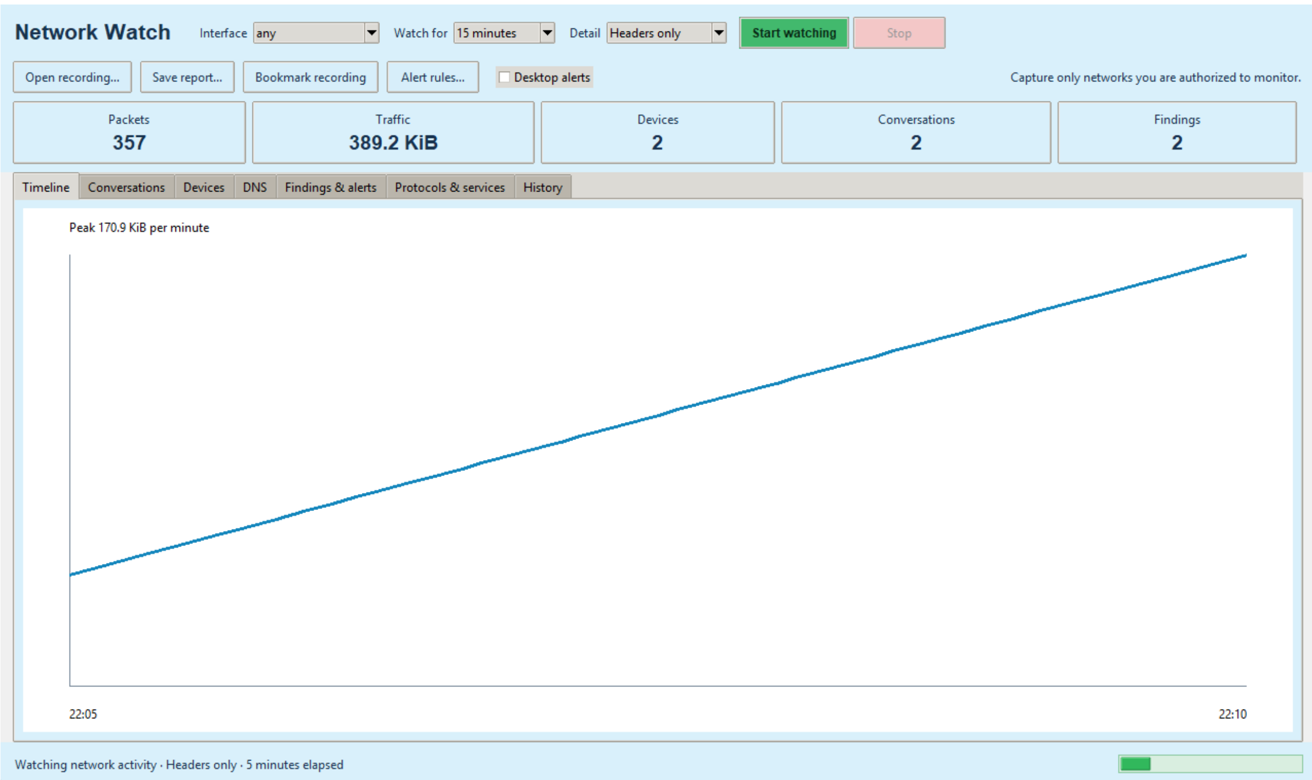

Figure: Network Watch dashboard

Open **Packets → Network Watch**, choose an interface, duration, and detail:

- Headers only: 128 bytes per packet; recommended.
- Protocol details: 512 bytes per packet.
- Full packets: up to complete packet payloads.

Confirm the authorized capture. The progress indicator runs while the watch is active and views refresh from the growing PCAP.

Dashboard tabs

packets/bytes, duration, and health.

protocols.

- **Timeline:** bytes per minute and peak traffic.
- **Conversations:** normalized bidirectional endpoints, ports, protocol,
- **Devices:** sent/received bytes, peers, external peers, observed ports, and
- **DNS:** query/response names, devices, servers, and response codes.
- **Findings & alerts:** severity, category, subject, and explanation.
- **Protocols & services:** protocol-layer and port totals.
- **History:** previous session time, duration, packets, bytes, and recording.

TCP health

Network Watch estimates resets, zero advertised windows, retransmissions, out-of-order sequences, unanswered SYNs, and SYN-to-SYN/ACK handshake time. Offloading, truncation, asymmetrical routing, or incomplete capture can affect these estimates; use them as investigative leads.

Findings and rules

Built-in findings include new saved-device baseline addresses, connection fan-out, large traffic increases, repeated DNS failure, resets, retransmissions, unanswered connections, ARP ownership changes, and unusually regular timing.

Use **Alert rules** to add:

- `new_device`;
- `traffic_bytes` threshold;
- `failed_connections` threshold;
- exact `destination` IP;
- observed `port`; or
- `dns_name` text.

Rules are bounded and stored atomically. Enable **Desktop alerts** to use `notify-send` when available. Duplicate notifications are suppressed per window.

Reports, bookmarks, retention

Save HTML, JSON, or conversation CSV reports. **Bookmark recording** copies the PCAP and a bounded note to a protected bookmark directory. Ordinary Network Watch captures and database history use seven-day retention; ordinary captures are also limited to 250 MiB, deleting oldest first. The recording being finalized and bookmarks are protected.

13. Passive Wi-Fi Watch

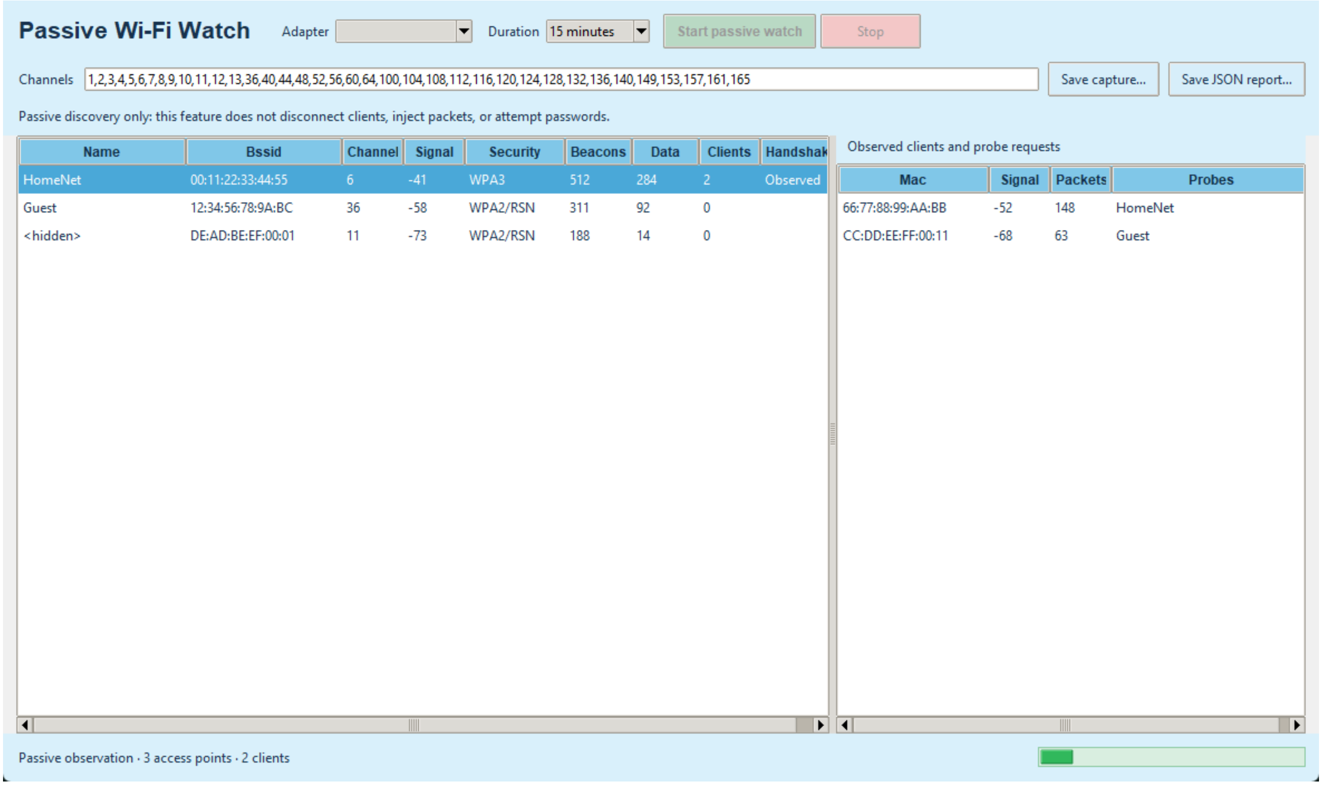

Figure: Passive Wi-Fi Watch

Install `iw`, attach a monitor-capable adapter, then open **Packets** → **Passive Wi-Fi Watch**. Choose the adapter, duration, and channel list. Confirm the legal scope and PolicyKit prompt.

The helper creates a temporary virtual interface named from the Linux interface index, sets it up, and changes only its validated channel. It does not stop NetworkManager, alter the managed interface MAC, or replace the base interface. On stop it removes only the reserved monitor interface after verifying its type.

The window displays:

- SSID/BSSID and hidden SSIDs;
- channel and radiotap dBm signal where supplied;
- advertised Open, WEP, WPA, WPA2/RSN, or WPA3 information;
- beacon/data counts;
- associated client MACs;
- passive probe-request names; and
- whether EAPOL traffic was observed.

“EAPOL observed” is not a recovered key or password. Save the radiotap PCAP or a JSON report if needed. Wi-Fi ordinary recordings use the same bounded retention.

Passive Wi-Fi Watch deliberately has no deauthentication, injection, rogue AP, credential interception, key/password cracking, or denial-of-service capability.

14. Automatic updates

The app checks the latest project GitHub release after startup. A newer valid tag causes the footer button to flash. Click it and confirm to:

1. download only the exact expected release asset URL;
2. enforce a 50 MiB package limit;

3. verify GitHub's SHA-256 digest;
4. verify Debian Package and Version fields;
5. launch the detached helper and close the GUI;
6. wait for the old process to end;
7. ask PolicyKit to run `apt-get install`; and
8. reopen the application.

Help contains a manual **Check for updates** action. Never bypass an update verification failure; use the Releases page and verify the checksum instead.

15. Command-line reference

Scan

```
advanced-ip-analyser scan TARGET [--timeout SECONDS] [--workers COUNT] [--output FILE]
```

Wake-on-LAN

```
advanced-ip-analyser wake MAC [--broadcast ADDRESS]
```

Capture

```
advanced-ip-analyser capture TARGET [--interface NAME] [--port PORT] \
  [--duration SECONDS] [--max-packets COUNT] [--output FILE]
advanced-ip-analyser capture-interfaces
```

Read/filter a capture

```
advanced-ip-analyser open-capture FILE [--host ADDRESS] [--port PORT] \
  [--limit COUNT] [--filter EXPRESSION]
```

Monitor and report

```
advanced-ip-analyser watch [--interface NAME] [--duration SECONDS] \
  [--snaplen BYTES] [--report FILE]
advanced-ip-analyser analyse-capture FILE --report FILE
```

16. Files, data, and privacy

Path	Contents
`~/.config/advanced-ip-analyser/favorites.json`	Favorites and notes
`~/.config/advanced-ip-analyser/packet-filters.json`	Named display filters
`~/.config/advanced-ip-analyser/alert-rules.json`	Network Watch rules
`~/.cache/advanced-ip-analyser/`	Short-lived selected-host captures and update downloads
`~/.local/share/advanced-ip-analyser/network-watch.sqlite3`	Session history and baselines
`~/.local/share/advanced-ip-analyser/captures/`	Ordinary Network Watch PCAP
`~/.local/share/advanced-ip-analyser/bookmarks/`	Bookmarked PCAP and note JSON
`~/.local/share/advanced-ip-analyser/wifi-captures/`	Passive Wi-Fi PCAP

The app has no analytics/telemetry endpoint. Local data leaves the computer only through explicit export/copy or tools the user launches. Update checks contact this project's GitHub Releases API.

17. Troubleshooting

Scan finds nothing

- Check the interface subnet and target spelling.
- Try Accurate and increase timeout.
- Confirm the target is powered and routed.
- A firewall may block ping and TCP probes.

Capture says the package must be installed

Privileged helpers are intentionally rejected from a source checkout because their root ownership cannot be trusted. Install the official `.deb`.

No Wi-Fi adapter appears

- Confirm the device exists under `/sys/class/net` and has a `wireless` directory.
- Install `iw`.
- Check adapter/driver monitor-mode and virtual-interface support.
- Some USB adapters need Debian firmware packages.

Channel hopping fails

Remove channels unsupported by the adapter, driver, or regulatory domain. The application does not override regulatory restrictions.

Service action fails

Install the corresponding Debian client/desktop handler. SSH also requires a valid remote username and normal remote authentication.

Remote power fails

Test `ssh -o BatchMode=yes user@host` separately. Configure key authentication and narrowly scoped passwordless `sudo shutdown` permission.

Update does not appear

Use Help → Check for updates. Confirm GitHub connectivity. A release without the expected filename or SHA-256 digest is deliberately ignored/rejected.

18. Keyboard and mouse reference

Input	Action
F5	Scan
Escape	Cancel scan
Ctrl+F	Focus result filter
Ctrl+O	Import inventory
Ctrl+S	Export inventory
Ctrl+Shift+C	Copy selected detail
Enter	Open selected supported service
Double-click	Open service/preferred web service
Right-click result	Open, copy, expand/collapse
Right-click TCP ports	Port presets

19. Security and interpretation notes

notifications, and Wi-Fi channel plans have explicit bounds.

- Scan/fingerprint work is bounded and unauthenticated.
- Network-derived strings are not passed through a shell.
- SSH option injection is rejected and `--` terminates options.
- XML entities/DOCTYPE are rejected and defused parsing is used.
- CSV formula-leading values are neutralized.
- HTML report/inventory values are escaped.
- Favorites, filters, and rules use atomic replacement.
- Packet files, records, filters, expressions, regex, reports, history, captures,
- Elevated helpers validate fixed options and filesystem ownership.
- Update packages require both cryptographic digest and Debian identity checks.

Network Watch findings and Passive Wi-Fi security labels are observations. They can be affected by partial visibility and should be confirmed with configuration, logs, and authorized specialist tools before action.

20. Getting help and contributing

Use the in-app Help centre for offline guidance. For bugs, include Debian version, application version, steps, expected/actual result, and sanitized logs/captures:

https://github.com/2E0LXY/Advanced-IP-Analyser/issues

Run tests before proposing source changes:

```
python3 -m unittest discover -s tests -v
python3 -m compileall -q src tests
```

The project is an independent GPL implementation and is not affiliated with Angry IP Scanner, Famatech Advanced IP Scanner, Airgorah, or Wireshark.